Lab: Robotic Systems Architecture and Markov Blankets

Objective

Analyze a robotic system by identifying its Markov blanket boundaries and mapping the sensor-actuator architecture to Active Inference components. You will decompose a mobile robot into internal states, sensory states, active states, and external states, then evaluate how this decomposition informs system design.

Prerequisites

- Familiarity with basic robot architectures (sensors, actuators, controllers)
- Understanding of the Markov blanket formalism (internal, external, sensory, active states)
- Basic block diagram or pseudocode skills

Part 1: System Boundary Identification

Consider a differential-drive mobile robot equipped with LIDAR, wheel encoders, an IMU, and two DC motors.

1. Draw or describe a block diagram that partitions the robot into Active Inference state types:
2. **Internal states**: onboard computation, belief states, generative model parameters
3. **Sensory states**: LIDAR range measurements, encoder ticks, IMU readings
4. **Active states**: motor voltage commands, LED indicators

5. **External states**: obstacles, floor surface, ambient lighting

6. For each sensory channel, specify the observation likelihood model p(o|s) in plain language. For example: "LIDAR returns a range measurement corrupted by Gaussian noise with variance proportional to distance."

Write your response here

Part 2: Generative Model Specification

Design a simplified generative model for the robot's position estimation:

1. Define the hidden state vector (e.g., x, y, theta).
2. Write the state transition model as a difference equation: $x_{t+1} = f(x_t, u_t) + w_t$.
3. Write the observation model for at least two sensor modalities: $z_t = h(x_t) + v_t$.
4. Specify reasonable noise covariance matrices Q (process) and R (observation).

Write your response here

Part 3: Sensor Fusion via Free Energy Minimization

Compare two approaches to fusing LIDAR and encoder data:

1. **Classical approach**: Describe how an Extended Kalman Filter would fuse these measurements.
2. **Active Inference approach**: Describe how variational free energy minimization over the generative model achieves the same fusion, emphasizing the role of precision weighting.
3. Under what conditions would the Active Inference approach weight LIDAR more heavily than encoders? When would it do the opposite?

Write your response here

Part 4: Architecture Trade-offs

Evaluate the following design decisions through the lens of Active Inference:

1. Adding a camera to the sensor suite: How does this change the generative model complexity? What is the trade-off between model accuracy and computational cost (model complexity term in free energy)?
2. Replacing DC motors with stepper motors: How does increased actuator precision affect the active states and the expected free energy of actions?
3. Moving computation from an onboard microcontroller to a cloud server: How does communication latency affect the temporal depth of the generative model?

Write your response here

Part 5: Pseudocode Design

Write pseudocode for a single perception-action cycle of your robot system:

```
function active_inference_step(belief, observation, model):
    # Step 1: Prediction error
    # Step 2: Belief update (perception)
    # Step 3: Action selection (minimize expected free energy)
    # Step 4: Execute action
    # Return updated belief
```

Fill in each step with operations specific to your differential-drive robot scenario.

Write your response here

Summary Table

Component	Classical Robotics Term	Active Inference Term	Your Robot Example
Sensors	Observation model	Likelihood mapping	
Actuators	Control input	Active states	
Controller	State estimator + planner	Free energy minimization	
World model	Dynamics model	Generative model	
Noise handling	Kalman gain	Precision weighting	

References

- Lanillos, P., et al. (2021). Active Inference in Robotics and Artificial Agents. *Frontiers in Neurorobotics*.
- Parr, T., Pezzulo, G., & Friston, K. J. (2022). *Active Inference*. MIT Press, Chapters 2-3.